

A Short Guide to Lemma C.1

Hybrid-measure fidelity and mass at infinity

Abstract

Lemma C.1 has two purposes. First, it extends the usual root fidelity of quantum states to classical–quantum states whose classical part is a Radon measure. Second, it shows that probability mass escaping to infinity during a weak-* limit has zero fidelity with a normalized target that remains in the original space. This note develops the background needed for those two claims and then explains the proof.

1 The picture before the formalism

For a finite classical register, a classical–quantum positive operator has the block-diagonal form

$$X = \bigoplus_i X_i, \quad Y = \bigoplus_i Y_i,$$

where the X_i, Y_i are positive trace-class operators. Orthogonal classical blocks do not interfere, so

$$B(X, Y) = \sum_i B(X_i, Y_i), \quad B(R, S) := \text{Tr} \sqrt{R^{1/2} S R^{1/2}}. \quad (1)$$

Lemma C.1 is the continuous version of (1): the sum over i becomes an integral over a classical parameter $y \in \mathbb{H}$.

The only additional difficulty is noncompactness. A sequence of probability measures may move farther and farther away and have no mass left on any fixed compact subset. Compactness is recovered by adjoining an abstract point at infinity. The second part of Lemma C.1 says that mass stored at this new point is orthogonal to a target supported on the original classical space.

2 Background

2.1 Compact and trace-class operators

Let $\mathcal{F}_{\text{orb}}$ be the separable multimode Fock space. We use

$$\mathcal{K}(\mathcal{F}_{\text{orb}}) = \{\text{compact operators on } \mathcal{F}_{\text{orb}}\}, \quad \mathcal{T}_1(\mathcal{F}_{\text{orb}}) = \{\text{trace-class operators on } \mathcal{F}_{\text{orb}}\}.$$

The trace class is the Banach dual of the compact operators, $\mathcal{K}(\mathcal{F}_{\text{orb}})^* = \mathcal{T}_1(\mathcal{F}_{\text{orb}})$, with pairing

$$\langle R, A \rangle = \text{Tr}(RA), \quad R \in \mathcal{T}_1(\mathcal{F}_{\text{orb}}), \quad A \in \mathcal{K}(\mathcal{F}_{\text{orb}}).$$

The mass of a positive element $R \in \mathcal{T}_1(\mathcal{F}_{\text{orb}})$ is $\text{Tr} R$; those with trace at most one are possibly subnormalized quantum states. If R and S are positive, the formula defining $B(R, S)$ above remains valid without any normalization assumption.

2.2 The algebra of continuous quantum-valued test functions

Let $\mathbf{H}$ be the finite-dimensional classical parameter space in the paper. The relevant nonunital C^* -algebra is

$$\mathfrak{A} = C_0(\mathbf{H}; \mathcal{K}(\mathcal{F}_{\text{orb}})).$$

Thus $A \in \mathfrak{A}$ is a norm-continuous compact-operator-valued function $y \mapsto A(y)$ which vanishes in operator norm as $\|y\| \rightarrow \infty$.

The vector-valued Riesz representation theorem identifies $C_0(X; E)^*$ with finite regular E^* -valued measures on X . Since $\mathcal{K}(\mathcal{F}_{\text{orb}})^* = \mathcal{T}_1(\mathcal{F}_{\text{orb}})$, every $\Xi \in \mathfrak{A}^*$ is therefore represented by a finite $\mathcal{T}_1(\mathcal{F}_{\text{orb}})$ -valued Radon measure through

$$\Xi(A) = \int_{\mathbf{H}} \text{Tr}[A(y) \Xi(\mathrm{d}y)].$$

If Ξ is a positive functional, then $\Xi(E) \geq 0$ for every Borel set E , so the representing measure is positive. Its scalar trace measure is

$$\mu_{\Xi}(E) := \text{Tr} \Xi(E).$$

If λ dominates μ_{Ξ} , the operator-valued Radon–Nikodym theorem gives a measurable density $X_y \in \mathcal{T}_1(\mathcal{F}_{\text{orb}})_+$ such that

$$\Xi(E) = \int_E X_y \, \mathrm{d}\lambda(y), \quad \Xi(A) = \int_{\mathbf{H}} \text{Tr}[X_y A(y)] \, \mathrm{d}\lambda(y).$$

Here one uses that $\mathcal{T}_1(\mathcal{F}_{\text{orb}})$ has the Radon–Nikodym property; separability of the Fock space is part of the standard setting. For two positive functionals Ξ and Υ , one may always take

$$\lambda = \mu_{\Xi} + \mu_{\Upsilon}.$$

They then have densities X_y and Y_y relative to the same scalar measure. This is the continuous analogue of writing two block-diagonal operators using the same list of classical blocks. These densities are positive trace-class weights, not necessarily normalized states. If $X_y = p_y \rho_y$ and $Y_y = q_y \sigma_y$, with ρ_y, σ_y normalized, then

$$\mathbf{B}(X_y, Y_y) = \sqrt{p_y q_y} \mathbf{B}(\rho_y, \sigma_y).$$

Thus the integrand combines the classical Hellinger overlap $\sqrt{p_y q_y}$ with the conditional quantum fidelity.

The Bochner space $L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}}))$ used in the main text embeds into $\mathfrak{A}^*$, but $\mathfrak{A}^*$ is larger: it also contains measures singular with respect to Lebesgue measure, such as point masses. This larger space is needed because weak limits of ordinary densities need not remain ordinary densities.

2.3 Transition amplitude

For density operators on a Hilbert space, $\mathbf{B}(R, S)$ is the root fidelity. Uhlmann’s theorem identifies it with the largest overlap between purifications of R and S . The Alberti–Uhlmann transition amplitude extends the same notion to arbitrary positive functionals on a C^* -algebra. Concretely,

$$\mathbf{B}_{\mathfrak{A}}(\phi, \psi) = \sup_{\pi, \xi, \eta} |\langle \xi, \eta \rangle|,$$

where π is a common representation and ξ, η implement ϕ, ψ in that representation. The facts needed here are:

- on trace-class operators it equals $\text{Tr } \sqrt{R^{1/2} S R^{1/2}}$;
- it is additive across orthogonal central direct summands;
- it is separately homogeneous: $B(aR, bS) = \sqrt{ab} B(R, S)$ for $a, b \geq 0$.

2.4 Unitization and the point at infinity

Because $\mathfrak{A}$ has no identity, its unitization is

$$\tilde{\mathfrak{A}} = \mathfrak{A} \oplus \mathbb{C}I.$$

Every element is written $A + tI$, and the quotient $\tilde{\mathfrak{A}}/\mathfrak{A} \cong \mathbb{C}$ has the character

$$\chi_\infty(A + tI) = t.$$

It is useful to regard χ_∞ as evaluation at an abstract point at infinity. If Υ is normalized on $\mathfrak{A}$, its canonical state extension is

$$\hat{\Upsilon}(A + tI) = \Upsilon(A) + t.$$

The bidual $\tilde{\mathfrak{A}}^{**}$ is a von Neumann algebra. The ideal $\mathfrak{A}$ has a central support projection $e \in \tilde{\mathfrak{A}}^{**}$. It separates the two mutually orthogonal parts

$$e\tilde{\mathfrak{A}}^{**} \quad (\text{the original space}), \quad (1 - e)\tilde{\mathfrak{A}}^{**} \quad (\text{the quotient at infinity}).$$

Restricting a state on $\tilde{\mathfrak{A}}$ to $\mathfrak{A}$ keeps its e -part and discards its $(1 - e)$ -part. The trace defect of the restriction is precisely the possible mass at infinity. For example, $e\hat{\Xi}$ denotes the functional $b \mapsto \hat{\Xi}(eb)$ on the e -summand.

3 Statement of Lemma C.1

Lemma C.1 (rephrased). *Let positive functionals $\Xi, \Upsilon \in \mathfrak{A}^*$ have densities X_y, Y_y relative to a common dominating measure λ . Then*

$$B(\Xi, \Upsilon) = \int_{\mathbf{H}} \text{Tr } \sqrt{X_y^{1/2} Y_y X_y^{1/2}} \, d\lambda(y). \quad (2)$$

The integral is independent of the chosen dominating measure and equals the Alberti–Uhlmann transition amplitude on $\mathfrak{A}$.

Now let $\hat{\Xi}$ be a state on $\tilde{\mathfrak{A}}$, let $\Xi = \hat{\Xi}|_{\mathfrak{A}}$, and let Υ be normalized on $\mathfrak{A}$. For the canonical extension $\hat{\Upsilon}$,

$$B_{\tilde{\mathfrak{A}}}(\hat{\Xi}, \hat{\Upsilon}) = B_{\mathfrak{A}}(\Xi, \Upsilon). \quad (3)$$

4 Why the fidelity integral is correct

4.1 Independence of the dominating measure

Suppose $\lambda = f\nu$. Densities relative to ν are $f(y)X_y$ and $f(y)Y_y$. Fiberwise homogeneity gives

$$B(f(y)X_y, f(y)Y_y) = f(y)B(X_y, Y_y).$$

Consequently,

$$\int B(fX, fY) \, d\nu = \int B(X, Y) \, d\lambda.$$

The same argument after passing to a common refinement handles any two dominating measures. Thus the right side of (2) depends only on Ξ and Υ .

4.2 The direct-integral argument

The Hilbert–Schmidt class is

$$\mathcal{T}_2(\mathcal{F}_{\text{orb}}) := \{Z \in \mathcal{B}(\mathcal{F}_{\text{orb}}) : \|Z\|_2^2 := \text{Tr}(Z^*Z) < \infty\}, \quad \langle Z, W \rangle_2 := \text{Tr}(Z^*W).$$

It is the standard Hilbert space for the algebra $\mathcal{K}(\mathcal{F}_{\text{orb}})$. Since $X_y^{1/2}$ and $Y_y^{1/2}$ are Hilbert–Schmidt, they define square-integrable fields in

$$\int_{\mathbf{H}}^{\oplus} \mathcal{T}_2(\mathcal{F}_{\text{orb}}) \, d\lambda(y).$$

More precisely, $\mathfrak{A}$ has the representation

$$(\pi(A)Z)_y = A(y)Z_y, \quad \|A(y)Z_y\|_2 \leq \|A(y)\| \|Z_y\|_2.$$

Thus “fiberwise” means that $A(y)$ acts by left multiplication at each y , without mixing different values of y . The square-root field $X^{1/2} : y \mapsto X_y^{1/2}$ implements Ξ since

$$\langle X^{1/2}, \pi(A)X^{1/2} \rangle = \int_{\mathbf{H}} \text{Tr}[X_y A(y)] \, d\lambda(y) = \Xi(A),$$

and the same identity with Y gives Υ .

Uhlmann’s theorem applied in each fiber says that the largest possible overlap at y is

$$\text{Tr} \sqrt{X_y^{1/2} Y_y X_y^{1/2}}.$$

The polar partial isometries can be chosen measurably as a function of y . After a fixed amplification if necessary (equivalently, by using measurable ε -maximizers), the pointwise choices form an admissible Uhlmann field. Integrating their overlaps proves that the transition amplitude is at least the right side of (2). Conversely, the fiberwise Uhlmann bound applies to every pair of implementing fields. More formally, $C_0(\mathbf{H}; \mathcal{K}(\mathcal{F}_{\text{orb}}))$ is type I, so every common representation disintegrates over $\mathbf{H}$ into multiples of the defining representation of $\mathcal{K}(\mathcal{F}_{\text{orb}})$. The local bound therefore controls every global overlap. This proves equality.

The argument is exactly the continuous form of the finite-block identity (1). The direct-integral language is needed only to justify that the fiberwise optimizations combine into one representation of the functionals.

5 Why mass at infinity contributes nothing

The central projection e gives orthogonal decompositions

$$\widehat{\Xi} = e\widehat{\Xi} + (1-e)\widehat{\Xi}, \quad \widehat{\Upsilon} = e\widehat{\Upsilon} + (1-e)\widehat{\Upsilon}.$$

Because Υ already has total mass one on $\mathfrak{A}$, its canonical extension has no mass left for the quotient:

$$(1-e)\widehat{\Upsilon} = 0.$$

Indeed, for an approximate identity (a_i) of $\mathfrak{A}$, one has $a_i \rightarrow e$ weak-* and $\widehat{\Upsilon}(e) = \lim_i \Upsilon(a_i) = \|\Upsilon\| = 1$. Additivity of transition amplitude over the two central summands now gives

$$\begin{aligned} B_{\widehat{\mathfrak{A}}}(\widehat{\Xi}, \widehat{\Upsilon}) &= B(e\widehat{\Xi}, e\widehat{\Upsilon}) + B((1-e)\widehat{\Xi}, 0) \\ &= B(e\widehat{\Xi}, e\widehat{\Upsilon}). \end{aligned}$$

The last term is precisely the transition amplitude of the restrictions Ξ and Υ to $\mathfrak{A}$: one uses $e\tilde{\mathfrak{A}}^{**} \cong \mathfrak{A}^{**}$ and invariance of transition amplitude under normal extension to the bidual. This proves (3).

Notice that the lemma does not say the escaping mass disappears from $\hat{\Xi}$. It remains a genuine component of that state, but it lies in a central sector where the target is zero, so its contribution to fidelity is zero.

A classical example

Take $\mathbf{H} = \mathbb{R}$ and ignore the quantum fiber. Let the target be the point mass $\Upsilon = \delta_0$, and consider on the one-point compactification

$$\hat{\Xi} = (1 - a)\delta_0 + a\delta_\infty, \quad 0 \leq a \leq 1.$$

Its restriction to $C_0(\mathbb{R})$ is $\Xi = (1 - a)\delta_0$. Classical root fidelity gives

$$B(\hat{\Xi}, \hat{\Upsilon}) = \sqrt{1 - a} = B(\Xi, \Upsilon).$$

The a units of mass at infinity pair with zero target mass and hence add nothing. Similarly, δ_n converges to δ_∞ in the unitization as $n \rightarrow \infty$, while its restriction converges vaguely to zero.

6 Why the paper needs Lemma C.1

In the Følner compactness argument following the lemma, averaged channels are extracted by weak-* compactness on the unitized algebra. A limiting output need not have an ordinary Bochner density: it may have a singular classical component, and some mass may escape to infinity. Lemma C.1 supplies the two facts needed to pass the fidelity payoff to that limit:

1. fidelity of the remaining classical–quantum measure is computed by the intrinsic fiberwise integral (2);
2. any quotient mass created at infinity has zero overlap with the normalized target, by (3).

This is why the lemma is formulated using positive functionals and Radon measures rather than only ordinary L^1 densities.

7 Minimal prerequisite checklist

To follow the first half of the lemma, it is enough to know root fidelity, Uhlmann’s theorem, Radon–Nikodym densities, and the finite classical–quantum block formula (1). The direct-integral formalism makes the continuous version rigorous.

The second half additionally uses three standard operator-algebra ideas: unitization of a nonunital C^* -algebra, the bidual as a von Neumann algebra, and decomposition by a central projection supporting an ideal. Its content is nevertheless simple: a target of total mass one in the original space has zero component at infinity, so an escaping component of the other state is orthogonal to it.